

COMPANION SLIDES

Chapter 3

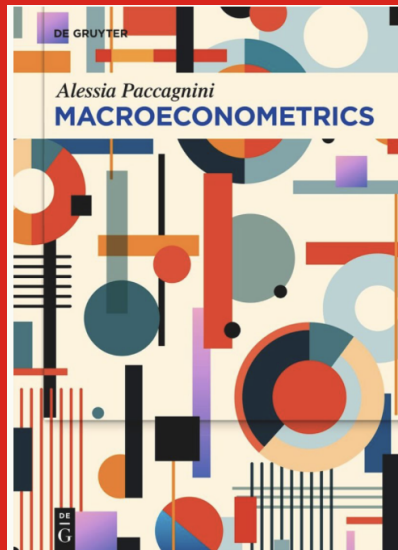
Local Projections

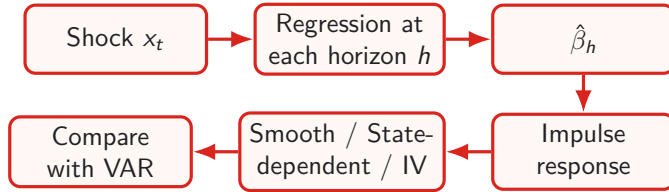
Estimate Impulse Responses Directly

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Key takeaway

Local projections estimate the dynamic response directly, one horizon at a time.

The VAR concern

- ▶ IRFs depend on the full dynamic system.
- ▶ Wrong lags, omitted variables or functional form can propagate misspecification across horizons.
- ▶ Recursive iteration can accumulate errors.

Jordà's alternative

For each horizon, regress the future outcome directly on the shock and controls. No recursive iteration is required.

Trade-off: robustness and flexibility versus lower efficiency, especially at long horizons.

- ▶ Write horizon-specific LP regressions.
- ▶ Interpret coefficients as impulse responses.
- ▶ Understand consistency and inference.
- ▶ Use lag augmentation and controls.
- ▶ Apply LP-IV for identification.
- ▶ Compare LP and VAR IRFs.
- ▶ Understand the bias-variance trade-off.
- ▶ Implement smooth LPs.
- ▶ Estimate state-dependent responses.
- ▶ Apply LPs to monetary and fiscal policy.

A separate regression for each horizon

$$y_{t+h} = \alpha_h + \beta_h x_t + \Gamma'_h w_t + u_{t+h}^{(h)}, \quad h = 0, 1, \dots, H.$$

- ▶ x_t is the shock or treatment variable.
- ▶ w_t collects controls and lags needed for identification and precision.
- ▶ β_h is the response of y_{t+h} to the shock at horizon h .
- ▶ Repeating the regression across horizons traces the IRF.

Key takeaway

The IRF is read directly from $\{\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_H\}$.

VAR route

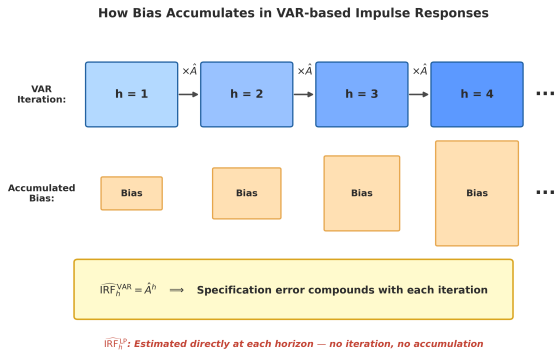
1. Estimate the dynamic system.
2. Identify the shock.
3. Iterate the system forward.
4. Extract the response.

LP route

1. Identify the shock.
2. Choose controls.
3. Regress y_{t+h} directly.
4. Repeat for every h .

Same target; different estimation strategy.

- ▶ In a VAR, small coefficient errors can compound as the model is iterated forward.
- ▶ LP avoids recursive propagation because each horizon is estimated directly.
- ▶ The price is noisier horizon-by-horizon estimates.



Why ordinary standard errors are not enough

For $h > 1$, adjacent LP observations use overlapping future outcomes. The regression errors are therefore serially correlated.

- ▶ Use heteroskedasticity- and autocorrelation-consistent inference where appropriate.
- ▶ In the monetary-policy example, the book uses Newey–West HAC standard errors with bandwidth $h + 1$.
- ▶ Confidence intervals typically widen at longer horizons as effective information falls.

Controls serve two roles

- ▶ support identification of the shock;
- ▶ improve precision by absorbing predictable dynamics.

Practical principle

Include a credible set of predetermined controls and sufficient lags, but remember that LP robustness does not remove the need for a defensible identification strategy.

When the shock is endogenous

Use an external instrument z_t for the endogenous shock x_t , then estimate horizon-specific IV regressions.

- ▶ The instrument must be relevant for the target shock.
- ▶ It must be exogenous to other structural disturbances affecting the outcome.
- ▶ LP-IV combines flexible dynamics with modern causal identification strategies.

Key takeaway

Local projections solve a dynamic specification problem; instruments solve an identification problem.

Under correct specification

VAR and LP estimators target the same population impulse responses and can be asymptotically equivalent under suitable conditions.

VAR advantage

- ▶ imposes cross-horizon dynamic structure
- ▶ often more efficient in finite samples

LP advantage

- ▶ less sensitive to dynamic misspecification
- ▶ naturally accommodates nonlinearities and flexible controls

VAR

Lower variance when the model is well specified, but potentially larger misspecification bias.

LP

Greater robustness to dynamic misspecification, but typically wider uncertainty bands, especially far out in the horizon.

Key takeaway

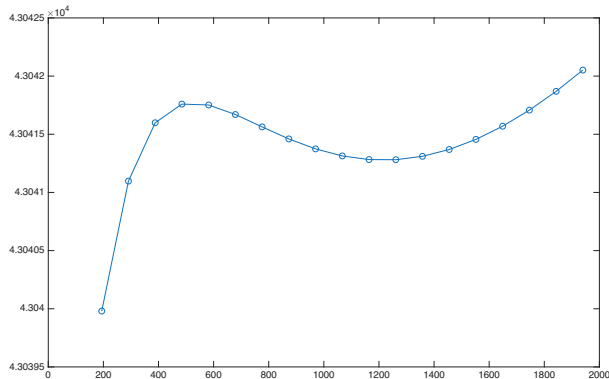
Method choice should reflect sample size, confidence in the dynamic specification, and the research question.

Borrow information across horizons

$$\beta(h) \approx \sum_{j=1}^J b_j B_j(h), \quad \hat{b} = \arg \min_b \left\{ RSS + \lambda \sum_{j=3}^J (b_j - 2b_{j-1} + b_{j-2})^2 \right\}.$$

- ▶ Cubic B-splines model the response path.
- ▶ The penalty discourages excessive curvature.
- ▶ $\lambda = 0$ gives no smoothing; larger λ produces smoother responses.

- ▶ The monetary-policy application selects λ by cross-validation.
- ▶ The book evaluates a grid scaled by effective sample size.
- ▶ The selected value balances fit and smoothness.



Responses can depend on the state of the economy

$$y_{t+h} = \alpha_h + \beta_h^{(0)}(1 - S_t)x_t + \beta_h^{(1)}S_tx_t + \Gamma_h'w_t + u_{t+h}^{(h)}.$$

- ▶ Compare recession versus expansion, high versus low uncertainty, or other regimes.
- ▶ Interactions make nonlinear extensions transparent.
- ▶ Interpretation requires care when the state itself is endogenous to the shock.

- ▶ Plot $\hat{\beta}_h$ against horizon h .
- ▶ Add uncertainty bands based on the chosen inference procedure.
- ▶ For cumulative responses, sum horizon-specific effects when economically appropriate.
- ▶ Structural interpretation still depends on identification – developed in Chapter 4.

Important distinction

LP is an estimator of dynamic responses. It is not, by itself, an identification scheme.

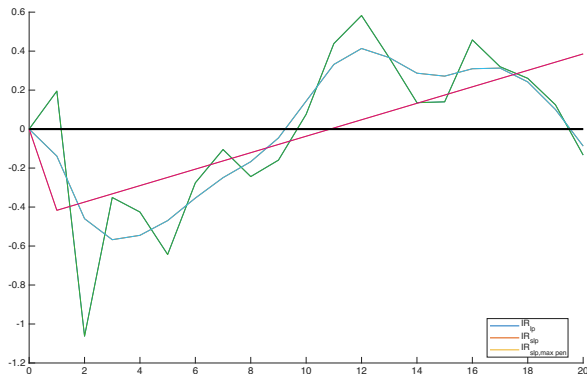
	Favor VAR	Favor LP
Dynamic specification	strong confidence	meaningful uncertainty
Sample size	limited	moderate / large
Efficiency	high priority	robustness high priority
Nonlinearities	model explicitly	easy interactions / states
Identification	available in VAR	IV / narrative / HFI integrate naturally
Transparency	system based	horizon-by-horizon

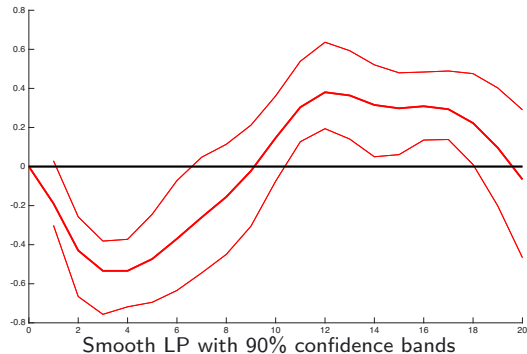
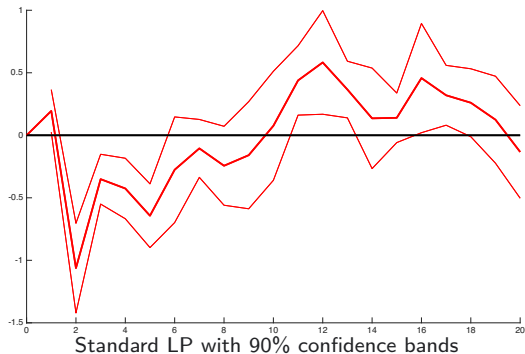
Smooth LP offers a useful middle ground when variance is a concern.

Sample and variables

- ▶ US quarterly data, 1959Q1–2007Q2.
- ▶ GDP growth, inflation, monetary-policy shock proxy.
- ▶ Four lags of the system.
- ▶ Horizons 1–20 quarters.

Goal: compare standard LP with smooth LP following Barnichon and Brownlees (2019).



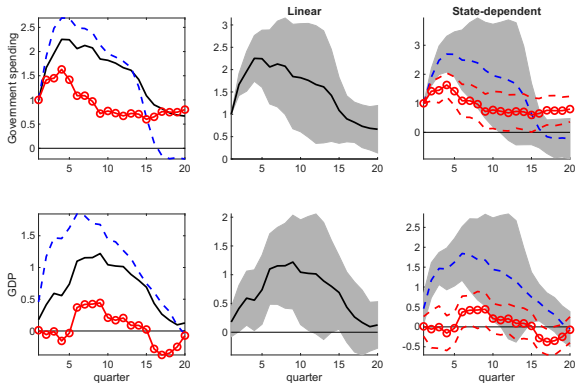


Key takeaway

Smoothing trades a controlled amount of bias for lower variance and a more interpretable response path.

Ramey–Zubairy application

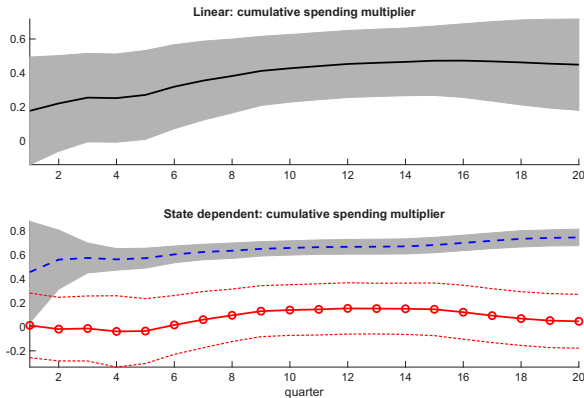
- ▶ historical US quarterly data, 1889–2015;
- ▶ government spending and GDP normalized by potential output;
- ▶ Blanchard–Perotti timing identification;
- ▶ cumulative multipliers from LP responses.



Economic object

How much cumulative output is generated by a cumulative increase in government spending over a given horizon?

- ▶ LP makes horizon-specific and cumulative effects transparent.
- ▶ State dependence can be added to ask whether multipliers differ across economic conditions.



1. LP estimates IRFs **directly** at each horizon.
2. LP is robust to dynamic misspecification but often less efficient than a correctly specified VAR.
3. Valid structural interpretation still requires a credible **identification strategy**.
4. Smooth and state-dependent LPs extend the framework without abandoning its transparency.
5. The best choice between VAR, LP and SLP depends on the application – not on a universal ranking.

Companion slides = starting point

These slides summarize core concepts and selected examples from Chapter 3. They are intentionally editable and are not a substitute for the full chapter.

- ▶ Add your own empirical applications and datasets.
- ▶ Add derivations or proofs for your audience.
- ▶ Add exercises, coding demonstrations and discussion questions.
- ▶ Replace or extend examples to match your course.

Make them yours.

Shock Identification

From dynamic responses to economically interpretable structural shocks

Next question

Once we can estimate a response, how do we know which structural shock generated it?